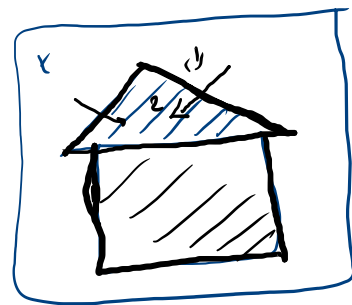
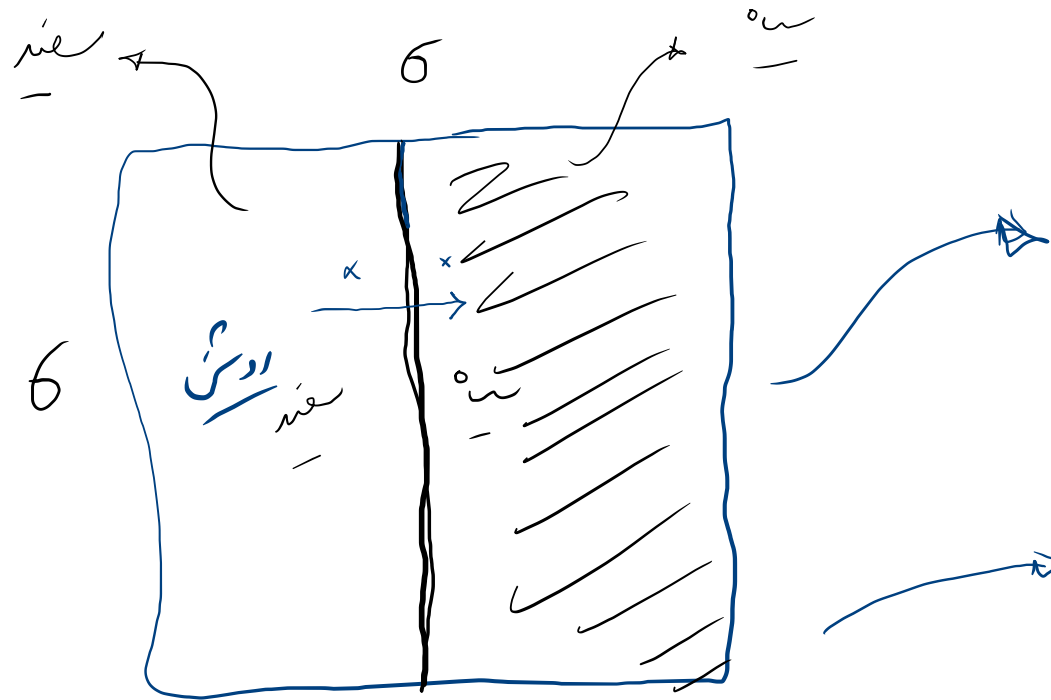


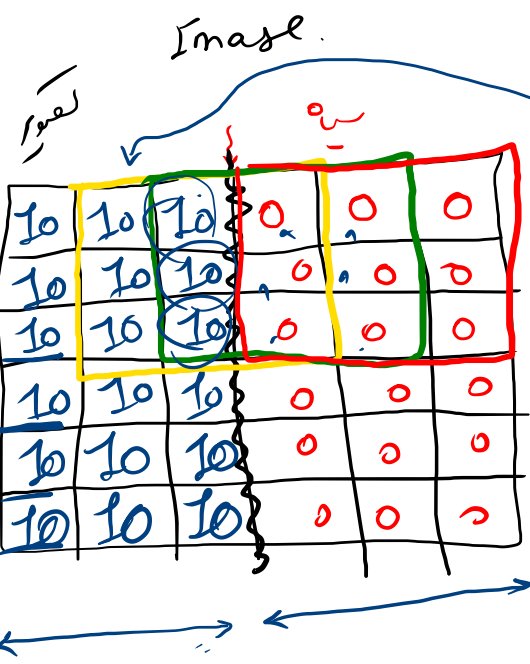
Convolutional  
neural  
nets.

2016 Jan

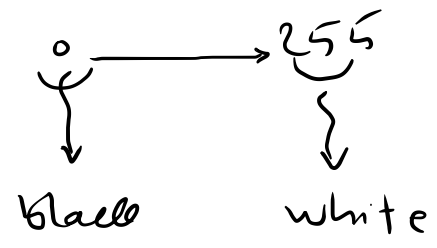


feature: edge / نی

آپری



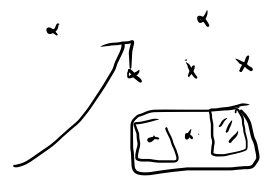
برای اینکه ما بتوانیم در این کسری را استخراج کنیم  
از آن نمودن استفاده می‌کنیم



|   |   |    |
|---|---|----|
| 1 | 0 | -1 |
| 1 | 0 | -1 |
| 1 | 0 | -1 |

=

$$\textcircled{1} : (1 \times 1) + (1 \times 0) + (1 \times -1) \\ (1 \times 1) + (1 \times 0) + (1 \times -1) \\ + (1 \times 1) + (1 \times 0) + (1 \times -1) = 0$$



$$I \otimes k \Rightarrow$$

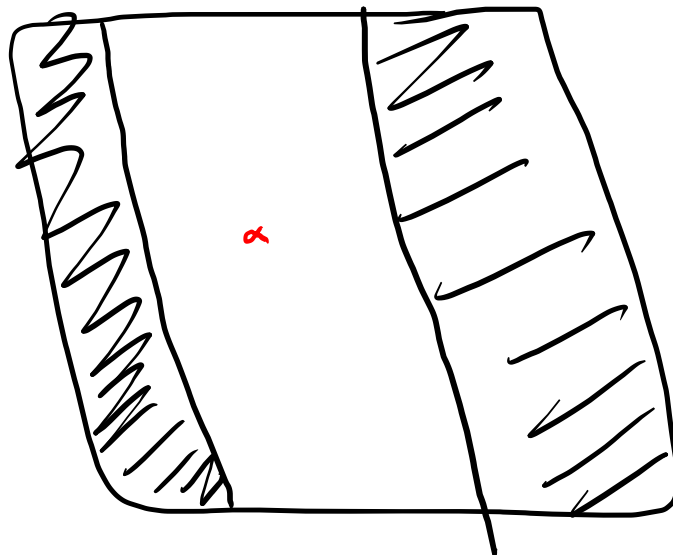
Kernel

|   |    |    |   |
|---|----|----|---|
| 0 | 30 | 30 | 0 |
| 0 | 30 | 30 | 0 |
| 0 | 30 | 30 | 0 |
| 0 | 30 | 30 | 0 |

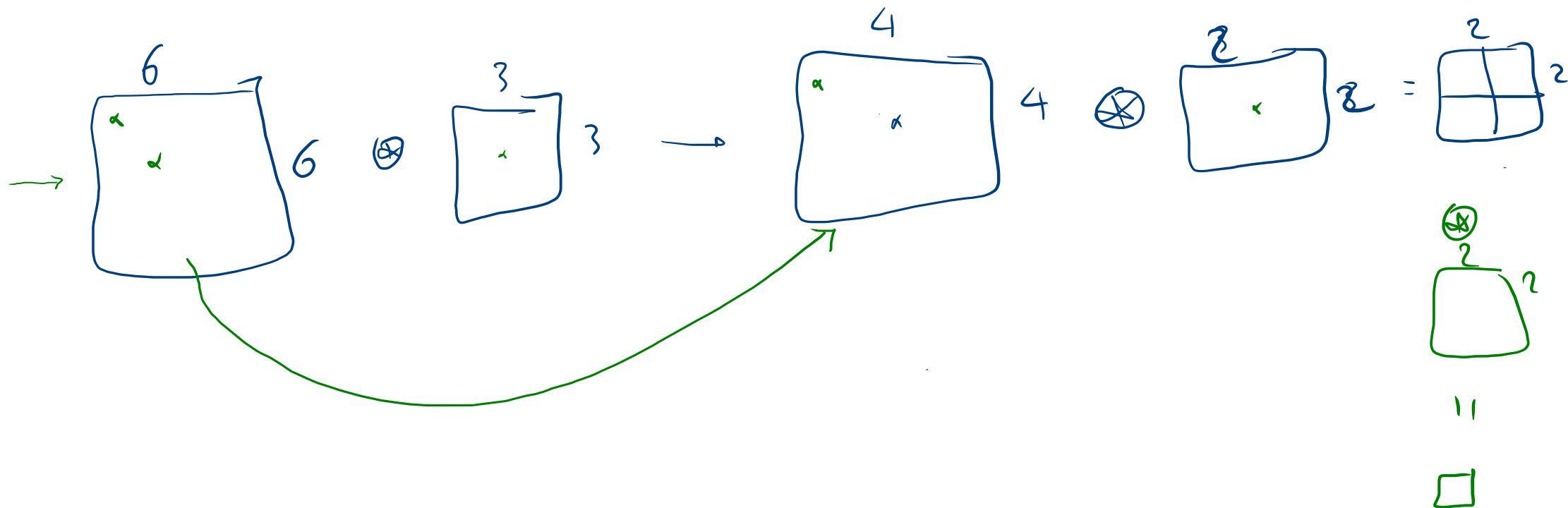
4

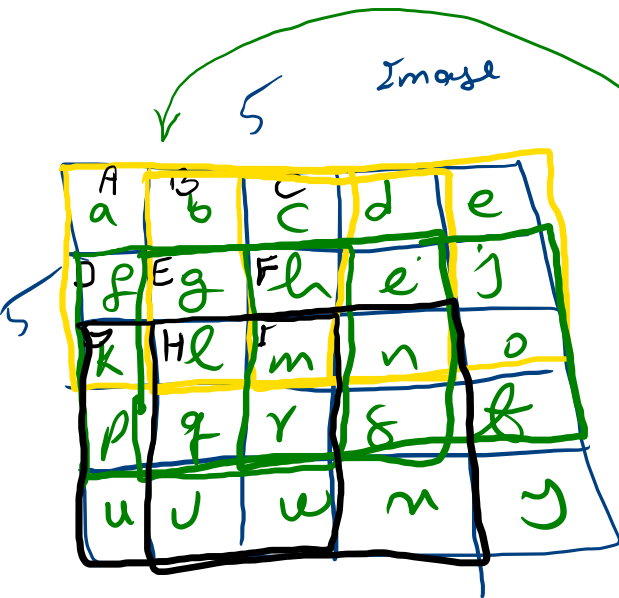
$$\textcircled{2} \quad (1 \times 1) + (0 \times 1) + (-1 \times 0) \\ (1 \times 1) + (0 \times 1) + (-1 \times 0) \\ (1 \times 1) + (0 \times 1) + (-1 \times 0) \\ = 30$$

|   |    |    |   |
|---|----|----|---|
| 0 | 30 | 30 | 0 |
| 0 | 30 | 30 | 0 |
| 0 | 30 | 30 | 0 |
| 0 | 30 | 30 | 0 |



$$\begin{array}{c} 6 \\ \boxed{\phantom{0000}} \end{array} \oplus \begin{array}{c} 3 \\ \boxed{\phantom{00}} \end{array} 3 = \begin{array}{c} \alpha \quad 4 \\ \boxed{\phantom{0000}} \end{array} 4$$





$I$

$K$

Diagram illustrating the convolution operation. The input image  $I$  (5x5 grid) is convolved with the kernel  $K$  (3x3 grid) to produce the output image  $O$  (3x3 grid). The output image is shown with the values of the convolution result.

|    |    |    |
|----|----|----|
| 1  | 7  | 12 |
| 7  | 15 | 14 |
| 15 | 17 | 9  |

$3 \times 3$

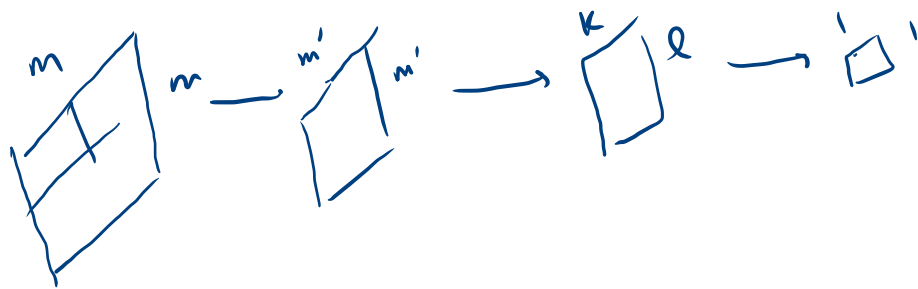
$$I \otimes K = \begin{bmatrix} (A \times a) + (B \times b) + (C \times c) + (D \times d) + (E \times e) + (F \times f) + (G \times g) + (H \times h) + (I \times i) \end{bmatrix}$$

$$\begin{aligned} (1) &: A \times b + B \times c + C \times d \\ &+ D \times e + E \times f + F \times g \\ &+ G \times h + H \times i + I \times j \end{aligned}$$

• از توکر شدن تعداد ریاضی سردی که توکوش سر شکر بزرگ محسوب می شود!

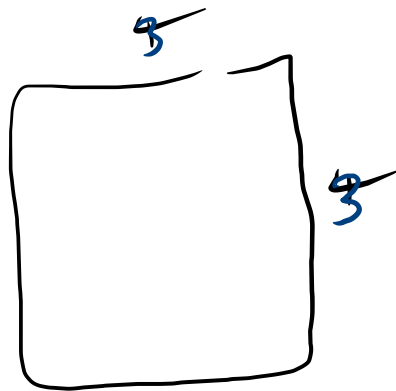
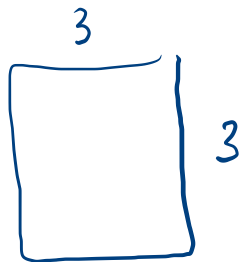
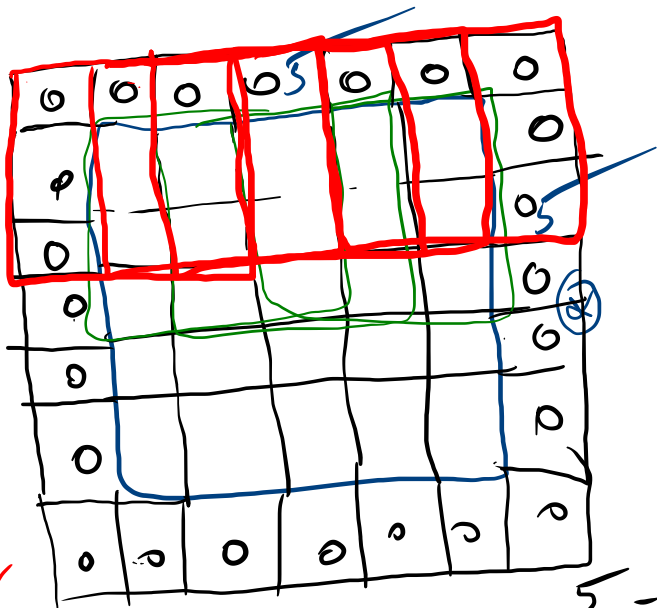
• اصل ریاضی است در در!

• توکر محله تعداد که ریاضی است تمام در ریاضی است!

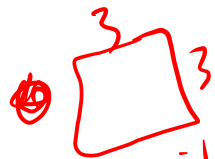


padding  $\rightarrow$  Zero padding!

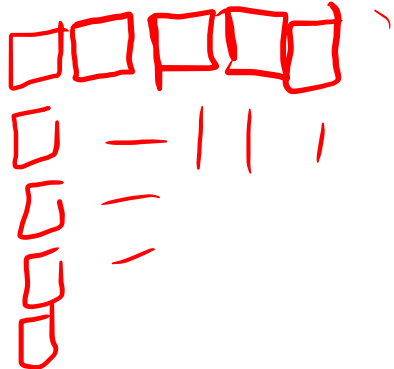




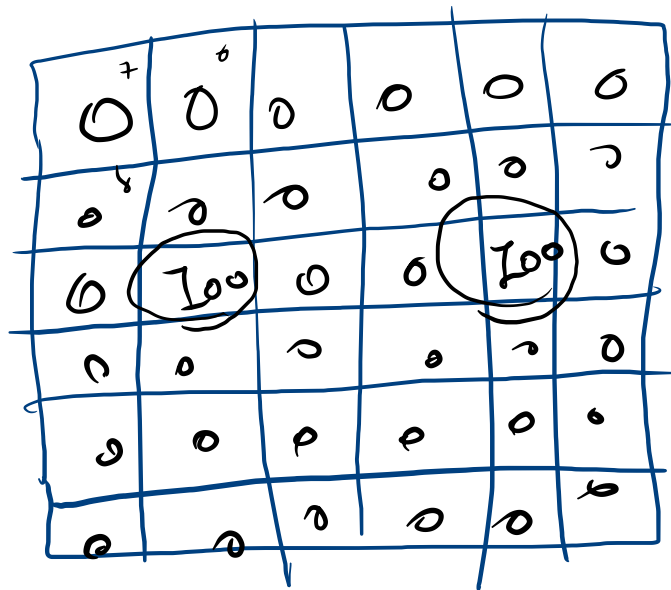
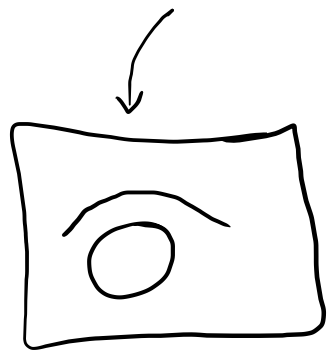
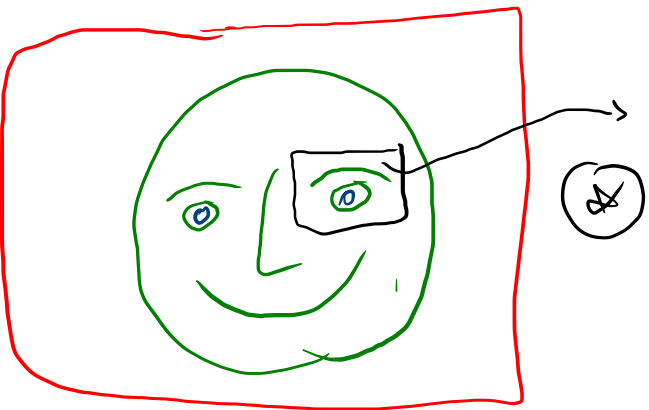
$$\frac{5 - 3}{2} = \frac{2}{2} = 1$$

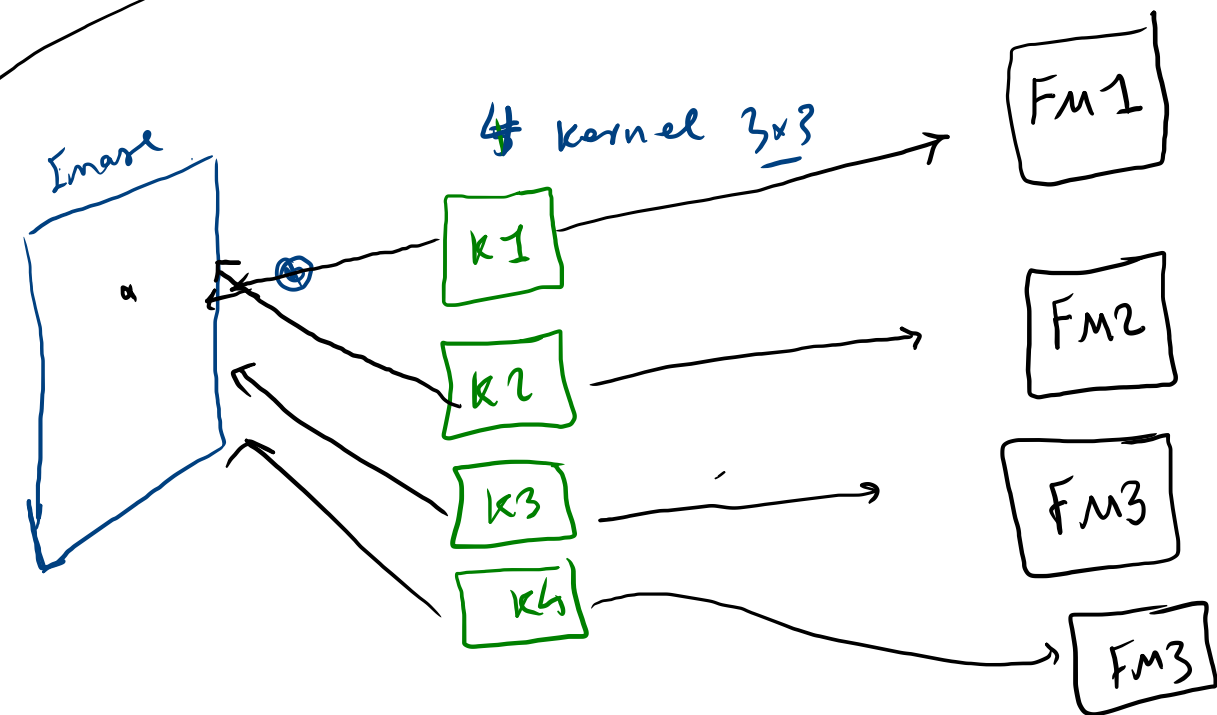
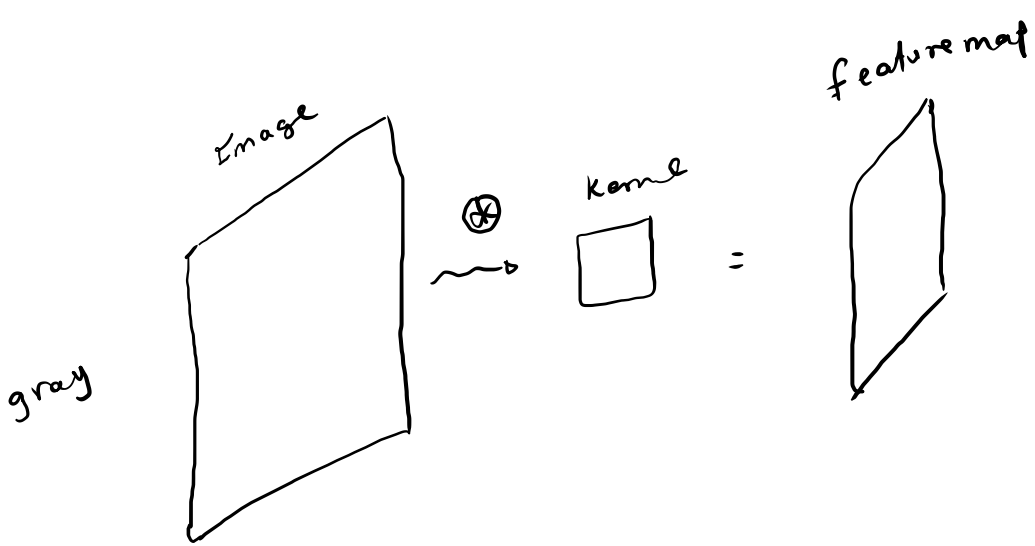


with padding



5x5



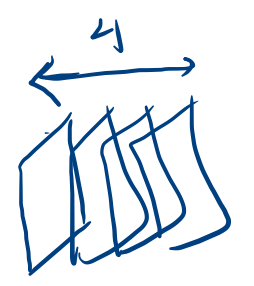


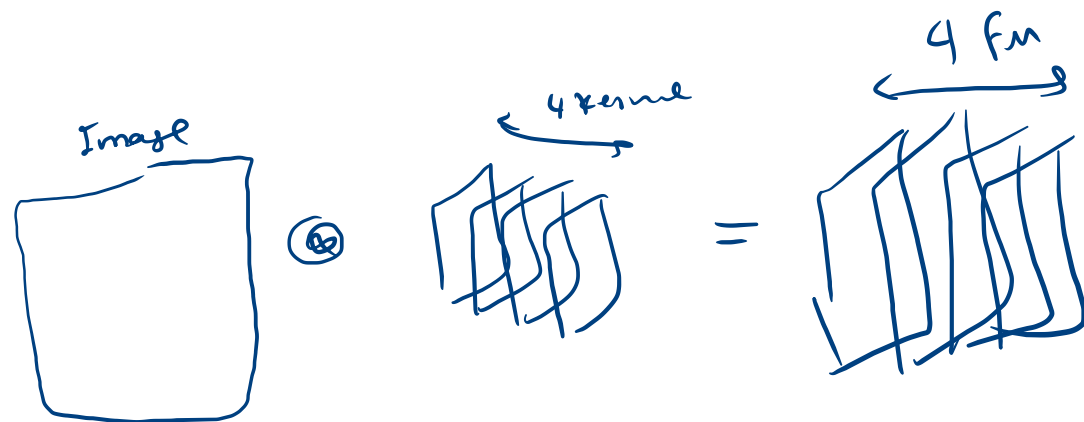
سبز رنگی

سبز آبی

نور

حجم

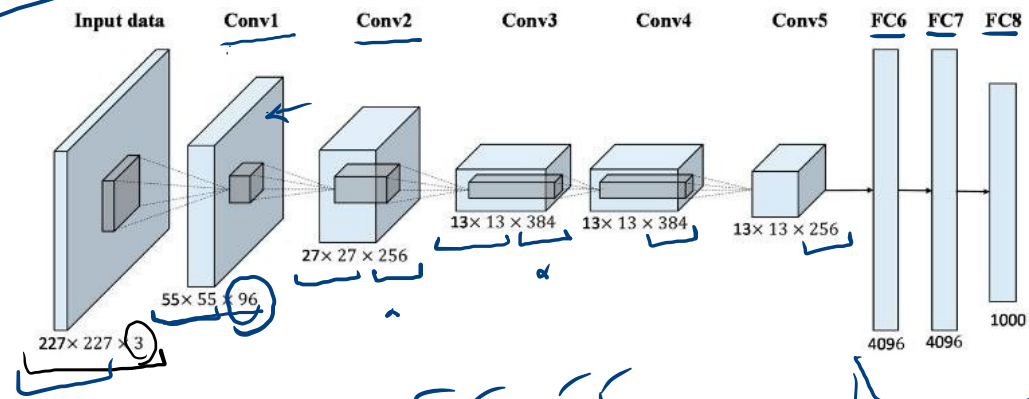






# Image classification

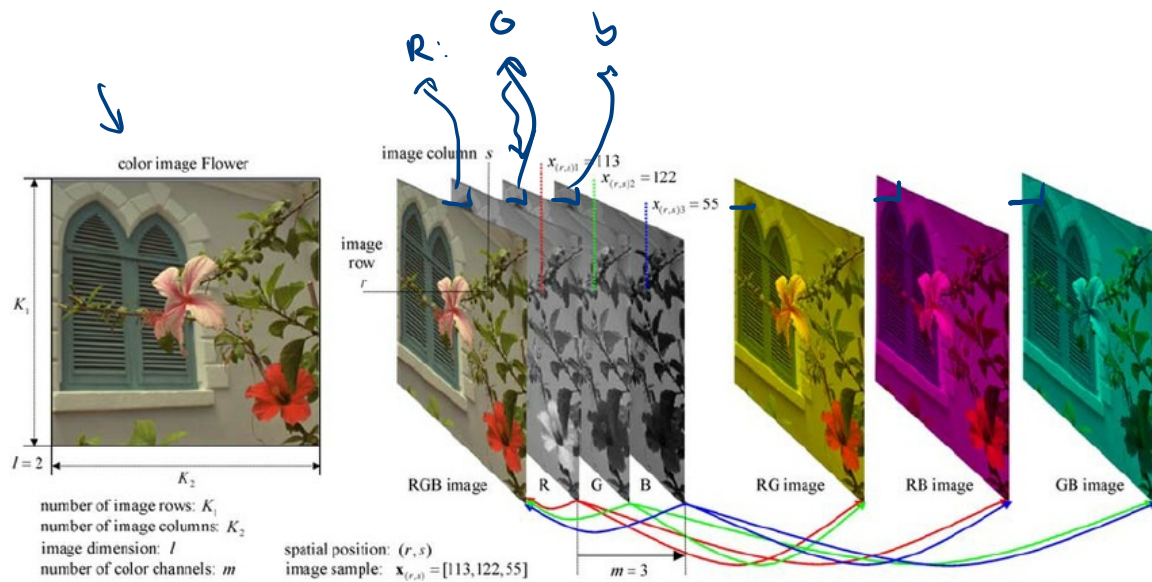
$\beta$   $\gamma$   $\delta$   
 $\alpha$   $\epsilon$   $\zeta$   
 $\eta$   $\theta$   $\iota$   
 $\kappa$   $\lambda$   $\mu$   
 $\nu$   $\xi$   $\omicron$   
 $\pi$   $\rho$   $\sigma$   
 $\tau$   $\upsilon$   $\phi$   
 $\chi$   $\psi$   $\omega$   
 $\alpha$   $\beta$   $\gamma$   
 $\delta$   $\epsilon$   $\zeta$   
 $\eta$   $\theta$   $\iota$   
 $\kappa$   $\lambda$   $\mu$   
 $\nu$   $\xi$   $\omicron$   
 $\pi$   $\rho$   $\sigma$   
 $\tau$   $\upsilon$   $\phi$   
 $\chi$   $\psi$   $\omega$



$227 \times 227 \rightarrow 55 \times 55$   
 $96$

Dense

Feature extractor



the 6rd

